

Deepak Mallampati

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Kent, OH | F-1 OPT, open to sponsorship

PROFESSIONAL SUMMARY

Applied AI engineer building agentic, multi-modal LLM products end-to-end - shipped a Chrome Web Store extension integrating four AI vision providers (Manga Lens), production agentic healthcare workflows with schema-validated tool contracts and evaluation audit trails (Progress Solutions), and an LLM-orchestrator-driven pixel-character generative system spanning Stable Diffusion + LoRA + ControlNet (Agentic Pixel). Comfortable across the stack - FastAPI + React/TS, vector DBs and embeddings, retries/fallbacks across model and API boundaries. Product-minded: cares as much about usability and creative control as technical novelty.

CORE COMPETENCIES

Agentic Orchestration • Multi-Modal Pipelines (Text · Image · Video) • Vector Memory + RAG • Multi-Provider AI Integration • Evaluation Frameworks • Structured Tool Use • Python • TypeScript • FastAPI • React • Diffusers • Stable Diffusion + LoRA + ControlNet

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present

Jr. AI/ML Engineer

USA - Healthcare Technology, AI Consulting

- Developed **agentic LLM workflows** for multi-step eligibility checks and care workflow navigation with **schema-validated structured outputs**, grounding rules, and evaluation audit trails; improved agent response stability **~25%**.
- Built **RAG systems** with recursive semantic chunking + transformer embeddings; **~35%** contextual retrieval precision gain, **>90%** retrieval-grounded alignment.
- Built **evaluation pipelines** for grounding, hallucination, and retrieval precision - automated checks + HITL scoring closing the loop on agent reliability.
- Packaged ML/LLM inference as **FastAPI/Flask** services, containerized with **Docker**; reduced post-deployment defects **~30%**.
- Shipped predictive ML pipelines (scikit-learn, XGBoost) for patient no-show prediction; **15-20%** recall gains on high-risk categories.

Suvidha Foundation

Jan 2022 - Mar 2022

Machine Learning Engineer

Hyderabad, India - Nonprofit Tech, Applied NLP

- Built a **transformer-based video summarization and highlight system** across 5,000+ sessions; **60-70%** manual review-time reduction with **~85%** alignment vs human curation.
- Implemented **document Q&A with RAG**; reduced hallucinated outputs **~30%**; deployed as **Flask API** with a lightweight UI for non-technical staff.

Energy Solutions International (Emerson)

Jun 2022 - Apr 2023

Database & DevOps Performance Engineer (Intern)

Hyderabad, India - Oil & Gas ERP

- Tuned T-SQL stored procedures; reduced query execution time **~20%** and data retrieval latency **~25%**.
- Built **Jenkins CI/CD pipelines** for SQL deployments; reduced deployment errors **>30%** and release cycle time **~35-40%**.

PROJECTS

Manga Lens - Chrome Web Store Extension, Real-Time AI Manga Translation shipped

Shipped multimodal Chrome extension integrating **four AI vision providers** (Claude Sonnet, GPT-4o mini, GPT-4.1 Nano, Ollama/minicpm-v local). Multi-section capture pipeline for tall webtoon panels with viewport-slice screenshots, coordinate remapping, dedup. Per-domain selector configs for 29 sites, 7-day translation cache, narrowed host permissions, provider-aware payload routing (WebP cloud / JPEG Ollama to avoid CUDA crash).

TypeScript · Manifest V3 · Multi-Provider Orchestration

Agentic Pixel Character Synthesis & Animation Engine

Phased generative AI system for identity-consistent, pose-controlled pixel character synthesis with sprite-sheet export. **Stable Diffusion** + **LoRA** for identity, **ControlNet** + **OpenPose/MediaPipe** for pose conditioning, latent-space consistency for animation smoothness, and an **LLM-based agent orchestrator** that decomposes high-level prompts into generation tasks.

PyTorch · Diffusers · FastAPI · LLM Orchestration

Dream Decoder - Multimodal Creative Intelligence Platform

FastAPI + React/TS/Vite/Tailwind app interpreting dreams, generating poetic reinterpretations, and synthesizing dreamscapes via coordinated multimodal APIs (Perplexity Sonar for interpretation, Replicate image models for 16:9 visuals). Intermediate **structured prompt transformation layers** improved contextual alignment **~30%** and first-pass image success rate **~25-30%**.

FastAPI · React · Multi-Modal API Orchestration

Agentic Healthcare Claims Processing & Fraud Risk Intelligence

Multi-agent pipeline with **schema-validated JSON contracts between agents** to prevent cascading hallucinations. RAG-grounded CPT/ICD validation against vector-indexed policy docs; ANN similarity for duplicate detection; explainable risk scoring with audit-ready reasoning traces.

Python · FastAPI · Vector Search · Structured LLM Tool Use

Driver Drowsiness Detection - Real-Time YOLOv8

Unified **YOLOv8** detection-and-classification replaced two-stage CNN; reduced inference latency **~30%**.

PyTorch · OpenCV · YOLOv8

EDUCATION

Master's Degree, Kent State University, Ohio, USA

Aug 2023 - May 2025

CERTIFICATIONS

- see Skills & Projects for portfolio depth

SKILLS

Languages & Frameworks: Python, TypeScript, FastAPI, Flask, React, Vite, Tailwind, SQL (PostgreSQL, T-SQL, SQLite), C++ (Arduino), PHP

LLM & GenAI: Agentic workflows, RAG, structured outputs, prompt engineering, evaluation pipelines, guardrails, grounding, embeddings, vector search, multi-provider orchestration (Claude, GPT, Ollama, Perplexity Sonar, Replicate)

Generative & Vision: Stable Diffusion, LoRA fine-tuning, ControlNet, OpenPose/MediaPipe, Hugging Face Diffusers, YOLOv8, OpenCV

ML Libraries: scikit-learn, XGBoost, PyTorch, Hugging Face Transformers, LangChain, LlamaIndex

Infra & DevOps: Docker, Jenkins, Grafana, CI/CD, RESTful APIs, observability, structured logging

Domains: Applied AI, Multi-Modal AI, Agentic Systems, Creative AI Tooling, Healthcare AI (HIPAA-conscious)